

trade-off between print speed and print quality. The quality of your print (product) depends on the size of the layers you use, but it also depends on the precision of the extruder (nozzle head). Heated build plate and precise nozzle alignment of the machine ensure high print quality. Printers with low layer thickness settings may take longer to produce a final print, but these generally produce a smoother finish. The best printers offer a wide range of quality settings, from fast (but low quality) to slow (but high quality).

Size of filament. All 3D printers have a unique set of temperature, speed and functional ranges in their respective designs, which you should review and consider when selecting filament. Most commonly used printing filaments are ABS, PLA, HIPS and PVA available in, generally, 1.75mm and 3mm diameter.

Availability of support. While buying a 3D printer, look for long periods of warranty. Most large 3D printers, especially those that come from big-name manufacturers, offer some type of specialised support. You can typically contact these companies for 'live' help and support over the phone, through email or live chat on their websites.

Sangani mentions, "Most of the buyers make their decision of buying a 3D printer based on cost. We highly recommend buyers to make their decision based on print quality and service support from the manufacturer."

Frahan Shaikh, founder, Mat-termatic, says, "The print time and quality can vary due to factors like layers and print speed. Fine print quality requires more layers in an object but it also increases the printing time."

Application areas

Your selection of the printer has to be based primarily on the use you

TABLE I
3D PRINTER PROCESS TECHNIQUES

Type	Technologies	Filament materials
Extrusion	FDM/FFF	Thermoplastics (ABS, PLA), eutectic metals, edible materials, rubbers, modelling clay, plasticine, metal clay
	DIW	Ceramic materials, metal alloy, cermet, metal matrix composite, ceramic matrix composite
Light polymerised	SLA	Photopolymer
	DLP	Photopolymer
Powder bed	3DP	Almost any metal alloy, powdered polymers, plaster
	EBM	Almost any metal alloy including titanium alloys
	SLM	Titanium alloys, cobalt chrome alloys, stainless steel, aluminium
	SHS	Thermoplastic powder
	SLS	Thermoplastics, metal powders, ceramic powders
	DMLS	Almost any metal alloy
Laminated	LOM	Paper, metal foil, plastic film
Wire	EBF	Almost any metal alloy

want to put it to. A printer useful for one type of application may not be useful for another type. See if the printer you are considering is for one of the following applications.

Industrial.

Automotive. To build tooling, jigs and fixtures from high-performance engineering thermoplastics using FDM technology

Construction. Super-size printers use special concrete and composite mixture that is thicker than regular concrete

Space and Defence. To print spare parts for planes or satellites using FDM technology

Healthcare. For implants and prosthetics, surgical guides, hearing aids and tissue engineering, etc

Computers. To make cases for laptops and other computers

Apparel. For clothing, consumer-grade eyewear, etc

Manufacturing.

Mass customisation. To customise 3D printed unique objects using simplified web based customisation software

Rapid manufacture. SLS and DMLS processes are enabling better rapid prototyping methods

Rapid prototyping. Rapid prototyping for research purposes using

powdered metals, casting media, plastics, paper or cartridges

Entertainment.

Film & Animation. For visual mock-ups, photo/film shoots and master patterns for moulding; ready for chrome or copper plating, PU finish, investment casting, etc

Trophies. Complex trophies can be made quickly with a surface finish that requires little or no hand finishing

Social.

Gifts. You can 3D print your child a toy, yourself a smartphone case, or your loved ones jewellery or chocolate/candies

Medicines. To produce your own medicine or household chemicals

Not all 3D printers come assembled; some of these come in DIY kits form, like the Makeblock Constructor 3D printer. This often results in price savings as the manufacturer does not need to assemble, calibrate and test the machine.

Open source robots are being built using 3D printers. Double Robotics grants access to their technology (an open SDK). On the other hand, 3&DBot is an Arduino 3D printer-robot with wheels and ODOI is a 3D printed humanoid robot.